Appendix

This is the Appendix for "Holtercare-Bench: A Multimodal Benchmark for Evaluating Long-Term Dynamic ECG Analysis".
This Appendix is organized as follows:

- Section A provides the detailed distribution and frequencies of the clinical annotations within Holtercare-23K.
- Section B presents the specific prompt templates utilized for LLM-based QA generation and report evaluation.
- Section C details supplemental generation metrics—*BLEU* n-gram overlap metrics, and additional analyses on modality alignment and statistical significance of fine-tuning improvements.

A Detailed Distribution of Clinical Annotations

To provide a deeper understanding of the scale and clinical diversity of Holtercare-23K, we present the exhaustive frequencies of our expert-verified beat and rhythm annotations.

Table 7 details the occurrence counts for all beat-level annotations. This includes a massive scale of valid QRS complexes—such as normal beats (N), atrial Fibrillation beats (Af), and Ventricular Premature beats (V)—alongside non-QRS elements and artifact.

Table 7: Frequency of beat-level annotations across Holtercare-23K.

Label	Description	Count
	Valid QRS Complexes	
N	Normal Beat	71,429,269
Af	Atrial Fibrillation	4,822,970
S	Atrial Premature Beat (APB)	1,042,095
P	Unclassified Pacing	840,298
V	Ventricular Premature Beat (VPB)	779,227
Se	Atrial Escape Beat	587,037
AF	Atrial Flutter	490,579
Je	Junctional Escape Beat	179,605
B	Bundle Branch Block	2,051
Ve	Ventricular Escape Beat	1,969
J	Junctional Premature Beat	182
?	Questionable Beat	30
F	Fusion Beat	1
aP	Atrial Single-Chamber Pacing	0
vP	Ventricular Single-Chamber Pacing	0
dP	Dual-Chamber Pacing	0
Ab	APB with Aberrant Ventricular Conduction	0
Va	Aberrant Ventricular Conduction	0
	Non-QRS Elements	
X	Artifact	833,824
Sa	Non-conducted APB	32,645
p	P-wave	0
t	T-wave	0

Table 8 outlines the comprehensive frequencies of the rhythm-level annotation. These capture complex, continuous cardiac events and diverse arrhythmias ranging from common Premature Atrial Contractions (PAC) to severe, life-threatening events like Ventricular Fibrillation (VF) and Asystole.

B Prompt for QA Generation and Report Evaluation

As outlined in our methodology, we use GPT-5-mini [37] for both the automated construction of complex reasoning tasks and the LLM-as-a-judge evaluation of long-context reports. To ensure transparency and reproducibility, we provide the exact representative prompts used in these two pipelines.

QA Generation. Figure 6 displays a representative structured prompt utilized by HolterAgent to generate *Open-QA* pairs, specifically showcasing the *Evidence Reasoning* sub-task. By feeding the model explicit patient information, the prompt strictly forces the LLM to extract timing, rhythm, and morphology clues directly from the provided text, effectively mitigating hallucinated waveform features.

Report Evaluation. Figure 5 presents a representative prompt design for our LLM-as-a-judge evaluation system, specifically showcasing the *General Summary* sub-task of *Report Generation.* Acting as a strict clinical expert, the prompt evaluates the generated output against the

Table 8: Frequency of rhythm-level annotations across Holtercare-23K.

Event	Count	Event	Count
Premature Atrial Contraction (PAC)	873	Premature Ventricular Contraction (PVC)	747
PAC Couplets	396	Atrial Tachycardia (AT)	339
PVC Couplets	152	Ventricular Tachycardia (VT)	90
PAC Bigeminy	82	PAC Trigeminy	64
PVC Trigeminy	58	PVC Bigeminy	55
Asystole	44	Sinus Arrhythmia	44
Long R-R Interval	33	Ventricular Fibrillation (VF)	23
Blocked PAC	15	Second-Degree Atrioventricular (AV) Block	13
First-Degree Atrioventricular (AV) Block	9	Accelerated Idioventricular Rhythm (AIVR)	6
Interpolated PVC	6	Ventricular Flutter (VFL)	6
Atrial Fibrillation (AF)	6	Atrial Flutter (AFL)	6
Sinus Arrest	6	Slow Junctional Escape Rhythm	5
Atrial Escape Rhythm	4	Junctional Escape Rhythm	4
Wandering Atrial Pacemaker (WAP)	4	Torsades de Pointes (TdP)	4
Accelerated Junctional Escape Rhythm	3	Ventricular Escape Rhythm	3
T-Wave Alternans (TWA)	2	Atrial Undersensing	2
Ventricular Capture Management (VCM)	2	Premature Junctional Contraction (PJC)	2
Atrial Capture Management (ACM)	2	Defibrillation	2
2:1 Atrioventricular (AV) Conduction	2	R-on-T PVC	2
ST Segment Changes	2	Accelerated Atrial Escape Rhythm	1
Anti-Tachycardia Pacing (ATP)	1	Ventricular Dissociation	1
OptiVol Fluid Status	1	Supraventricular Tachycardia (SVT)	1
Right Ventricular Capture Management (RVCM)	1	Ventricular Sense Response (VSR)	1
Chest Compression Waveform	1		

ground truth reference across 17 granular criteria. The specific fine-grained criteria evaluated by the LLM judge are detailed previously in Table 2 for *Statistical Overview* and Table 3 for *General Summary*. The prompt is explicitly designed to severely deduct points for fabricated information, missing values, or hallucinated clinical diagnoses.

C Supplemental Experimental Results

C.1 Supplemental Generation Metrics

specifically the *BLEU* n-gram overlap metrics for *Open-QA* and *Report Generation* tasks

While the main manuscript primarily focuses on clinical accuracy, specialized semantic metrics like *F1-Bio*, and automated LLM judge *Score*^{GPT} to assess reasoning capabilities, standard *BLEU* [39] n-gram overlap metrics provide a useful supplementary perspective on textual generation quality.

Table 9 presents *BLEU-1* and *BLEU-4* scores achieved by the evaluated baseline and fine-tuned models on *Open-QA* tasks, specifically evaluating *Event Timing*, *Diagnosis*, and *Evidence Reasoning* sub-tasks.

Table 10 details the corresponding *BLEU-1* and *BLEU-4* metrics for *Report Generation* tasks, covering both *Statistical Overview* and *General Summary*. These supplemental metrics highlight the lexical and structural alignment between the models' generated responses and the expert-crafted ground truth.

C.2 Modality Alignment and Representation Analysis

To accommodate the diverse architectural constraints of contemporary MLLMs, our data engine, HolterAgent, constructs a format compatibility pipeline that transforms a single, unified .edf signal source into three distinct representation formats: raw signal, video stream, and clinical text. This design is fundamentally intended to maximize MLLM compatibility rather than to introduce three independent data sources. Specifically, the video modality is prioritized for its ability to preserve the continuous temporal dynamics and morphological evolution of streaming electrophysiological data. Conversely, the text modality serves as a robust fallback for models lacking native video processing capabilities, representing the signal through discretized numerical sequences.

To empirically verify that this multi-representation design does not introduce modality-specific artifacts or bias the evaluation, we conducted a comprehensive comparative analysis. Table 11 presents the *Closed-QA* performance of all video-capable models when evaluated exclusively under the text modality. The results demonstrate that neither modality consistently dominates across all models and tasks. For

Report Evaluation Prompt

You are a strict and professional cardiac expert grading an LLM-generated Holter ECG report against a ground truth reference. Actively look for missing numbers, wrong values, fabricated information, and formatting issues. Deduct points severely for any discrepancy. **Do not default to 100.**

Reference Report: `{{REFERENCE_REPORT}}`
Generated Report: `{{GENERATED_REPORT}}`

Evaluation Criteria

(1) Accurate extraction of global metrics
(2) Accurate reporting of HR extremes
(3) Correct extraction of tachycardia and bradycardia burdens
(4) Correct classification of the baseline rhythm (e.g., AFib)
(5) Accurate total count and burden of PACs/PVCs
(6) Precise breakdown of ectopic patterns (e.g., isolated, paired, runs)
(7) Complete inclusion of severe events (e.g., VT runs, VF, asystole)
(8) Correct interpretation of conduction blocks or pacing signals
(9) Correct reporting of ST-segment and T-wave changes
(10) Strict consistency with clinical facts without hallucinated findings
(11) Objective description without exaggeration or understatement
(12) Clinical diagnoses strictly grounded in supporting statistical data
(13) Coherent narrative structure without fragmented data enumeration
(14) Structured clinical hierarchy, prioritizing major diagnoses over secondary findings
(15) Precise application of medical terminology
(16) Consistent numeric formatting and standardized unit application (e.g., "bpm" for rate, "%" for burden)
(17) Inclusion of appropriate clinical caveats

Requirements

(1) Score each item from 0 to 100 based on the degree of compliance with the specific criterion.
- **Perfect (90–100):** Perfect or near-perfect compliance with this specific criterion.
- **Substantial (70–89):** Substantial compliance, with minor flaws, slight inaccuracies, or trivial omissions.
- **Partial (40–69):** Captures the relevant clinical concept, but applies an incorrect metric type or statistical aggregation.
- **Poor (10–39):** Barely related or severely flawed, but not completely blank.
- **Failure (0–9):** Complete failure, the concept is entirely missing, or severe hallucination.

(2) If a specific clinical finding (e.g., ectopic patterns, severe events, ST-segment changes) is not mentioned in the reference report, it means the patient does not have it.
- If the generated report correctly omits it as well, this is a perfect match for that criterion. **Do not deduct points for missing a condition that isn't in the reference.**
- However, if the reference report does not have it, but the generated report fabricates or hallucinates it, you must severely deduct points.

(3) The output must be a valid JSON object where keys are the item numbers ("1" to "17") and values are purely numerical scores from 0 to 100. **Do not include any other text.** For example:

```
{
    "1": 95, "2": 80, ..., "17": 100
}
```

Figure 5: Prompt for report evaluation in *General Summary* tasks of *Report Generation.*

instance, while certain models exhibit marginal improvements in specific tasks under the text modality, others show a clear preference for video inputs. This non-uniform performance distribution confirms that the observed performance gaps among different MLLMs reflect their intrinsic architectural capabilities and inductive biases in processing long-term physiological sequences, rather than artifacts induced by our data formatting pipeline.

QA Generation Prompt

Your task is to generate high-fidelity *Open-QA* data in *Evidence Reasoning* tasks for Holter ECG diagnostic training. Given a patient's structured information, you must generate exactly one QA pair, construct a rigorous, evidence-based clinical reasoning process that verifies the target diagnosis based solely on the provided inputs. Zero hallucination is tolerated. Adhere strictly to requirements below.

Patient Information

- **Age:** `{{AGE}}`
- **Gender:** `{{GENDER}}`
- **Electronic Medical Records:** `{{EMR}}`
- **Beat-Level Annotations:** `{{BEAT_ANNOTATIONS}}`
- **Rhythm-Level Annotations:** `{{RHYTHM_ANNOTATIONS}}`
- **Report-Level Summaries:** `{{REPORT_SUMMARY}}`
- **Target Disease**: `{{TARGET_DISEASE}}`

Requirements

(1) Requirements for **questions**:
- Introduce the patient's **age** and **gender** (e.g., *"The current monitoring segment comes from a 61-year-old male."*), explicitly mention the exact **target disease**, and ask whether it is present on the visible ECG waveform.
- Ask the reader to specify which waveform features support that determination (e.g., prompting for rhythm clues, timing, P-wave morphology, PR-interval behavior, or QRS characteristics).
- Vary the phrasing and sentence structure naturally, but **do not change the core verification task.**

(2) Requirements for **answers**:
- Begin by directly confirming the presence of the target disease (e.g., *"Yes, {{TARGET_DISEASE}} is present."* or *"Confirmed."*).
- Provide a structured clinical reasoning process by summarizing clues extracted from the provided annotations and summaries. Organize the reasoning into logical clinical dimensions where applicable, such as:
  - **Timing / Rhythm Clues:** Discuss rates, RR intervals, coupling intervals, and event burden based on the Report and Beat/Rhythm Annotations.
  - **Morphology Clues:** Describe wave shapes (P, QRS, T, ST-segment deviations) and AV conduction characteristics relevant to the disease.
- Base all explanations solely on the provided information. **Do not hallucinate waveform features that cannot be inferred or documented from these inputs.**

(3) The output must be strictly in the form of JSON:

```
{
    "question": "...",
    "answer": "..."
}
```

Figure 6: Prompt for QA pairs generation in *Evidence Reasoning* tasks of *Open-QA*.

C.3 Statistical Significance Analysis

To rigorously confirm whether the substantial performance improvements observed after fine-tuning are statistically robust and not attributable to random sampling variance or favorable test-set splits, we performed McNemar's tests on all five *Closed-QA* tasks for both fine-tuned representative models, Phi-4-mini-3.8B [1] and Qwen3-VL-8B [3]. As detailed in Table 12, all pairwise comparisons between the zero-shot baseline and fine-tuned models yield p-values $\ll 0.001$. These consistently extreme statistical significances across all tasks and both models provide overwhelming evidence that the fine-tuning process induces genuine capability acquisition and knowledge internalization in long-context ECG reasoning, rather than mere fluctuations in sampling variance.

Table 9: Supplementary performance comparison of evaluated models on *Open-QA* tasks from Holtercare-Bench, evaluated by additional *BLEU* n-gram overlap metrics.

Model	Modality	Event Timing		Diagnosis		Evidence Reasoning	
		BLEU-1 ↑	BLEU-4 ↑	BLEU-1 ↑	BLEU-4 ↑	BLEU-1 ↑	BLEU-4 ↑
Generalist Models							
GPT-5-mini	A	37.50	10.83	8.70	1.96	96.67	38.48
Claude-4.5-Haiku	A	11.68	1.25	6.98	0.55	57.53	8.97
Phi-4-mini-3.8B	A	53.57	13.22	53.85	15.73	96.49	55.56
Phi-4-mini-3.8B^{FT}	A	**95.00**	**77.39**	**98.46**	**87.05**	**93.70**	**72.64**
InternVL-3.5-8B	🎞	41.46	8.65	35.29	6.02	96.30	30.39
MiniCPM-V4.5-8B	🎞	16.33	2.06	15.79	1.25	91.49	35.50
Gemini-3.0-Flash	🎞	66.67	22.93	17.65	2.87	96.88	45.06
Qwen3-VL-8B	🎞	56.67	12.68	50.00	0.00	95.35	52.34
Qwen3-VL-8B^{FT}	🎞	60.95	31.91	30.77	12.36	97.14	50.30
Medical Models							
LLaVA-Med-V1.5-7B	A	37.25	7.50	37.50	5.45	86.05	42.02
MedGemma-1.5-4B-IT	A	22.22	2.90	16.00	2.13	97.62	65.55
HealthGPT-M3-3.8B	A	34.21	6.19	12.36	1.24	90.24	43.46
Lingshu-7B	🎞	33.33	5.61	25.00	3.98	97.67	47.90
MedVLM-R1-2B	🎞	40.54	6.60	33.33	4.50	85.96	23.23
HuatuoGPT-Vision-7B	🎞	53.33	21.02	33.33	8.05	98.32	8.16

Table 10: Supplementary performance comparison of evaluated models on *Report Generation* tasks from Holtercare-Bench, evaluated by additional *BLEU* n-gram overlap metrics.

Model	Modality	Statistical Overview		General Summary	
		BLEU-1 ↑	BLEU-4 ↑	BLEU-1 ↑	BLEU-4 ↑
Generalist Models					
GPT-5-mini	A	44.64	3.65	60.47	6.95
Claude-4.5-Haiku	A	15.92	1.15	27.60	1.00
Phi-4-mini-3.8B	A	56.76	10.99	52.43	6.03
Phi-4-mini-3.8B^{FT}	A	**97.67**	**68.03**	**94.83**	**71.94**
InternVL-3.5-8B	🎞	81.63	42.20	71.43	14.88
MiniCPM-V4.5-8B	🎞	42.55	6.95	61.96	12.88
Gemini-3.0-Flash	🎞	51.11	5.48	67.07	9.86
Qwen3-VL-8B	🎞	34.75	8.53	41.13	3.22
Qwen3-VL-8B^{FT}	🎞	47.31	4.88	50.56	4.73
Medical Models					
LLaVA-Med-V1.5-7B	A	45.76	3.53	58.73	8.09
MedGemma-1.5-4B-IT	A	59.62	9.42	46.81	3.94
HealthGPT-M3-3.8B	A	10.79	1.47	52.63	10.27
Lingshu-7B	🎞	38.82	7.56	59.82	10.40
MedVLM-R1-2B	🎞	68.29	18.42	60.00	7.18
HuatuoGPT-Vision-7B	🎞	37.63	4.73	52.14	7.05

Table 11: Supplementary performance comparison of video-capable models evaluated under the text modality on *Closed-QA* tasks from Holtercare-Bench, evaluated by accuracy.

Model	Modality	Presence	Event Counting	Event Timing	HR Extremum Timing	Diagnosis
Generalist Models						
InternVL-3.5-8B	A	58.01	29.66	47.26	31.85	39.37
MiniCPM-V4.5-8B	A	54.90	51.38	28.05	35.99	26.38
Gemini-3.0-Flash	A	67.32	14.98	62.50	53.18	41.73
Qwen3-VL-8B	A	60.95	31.80	28.66	34.71	32.28
Qwen3-VL-8B^{FT}	A	95.10	83.79	97.23	71.97	94.88
Medical Models						
Lingshu-7B	A	47.55	35.17	24.70	29.94	25.98
MedVLM-R1-2B	A	40.20	41.28	26.52	32.17	13.39
HuatuoGPT-Vision-7B	A	52.29	25.38	26.22	32.17	23.23

Table 12: Statistical significance analysis of improvements from zero-shot baselines to fine-tuned models on *Closed-QA* tasks from Holtercare-Bench. The reported p-values are derived from McNemar tests.

Model	Modality	Presence	Event Counting	Event Timing	HR Extremum Timing	Diagnosis
Phi-4-mini-3.8B	A	1.95×10^{-5}	3.99×10^{-16}	6.33×10^{-20}	3.09×10^{-8}	1.76×10^{-11}
Qwen3-VL-8B	A	6.17×10^{-41}	5.26×10^{-32}	2.18×10^{-52}	6.57×10^{-18}	3.55×10^{-35}
Qwen3-VL-8B	🎞	7.08×10^{-44}	3.14×10^{-36}	6.14×10^{-47}	1.08×10^{-17}	1.30×10^{-28}